

Scalability & Future Plan

Date 29 September 2026
Team ID _____

Project Name FitBuddy – AI Fitness Plan Generator using Gemini Models

Maximum Marks 1 Mark

Scalability & Future Plan:

This document outlines how the current FitBuddy solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's proposed roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Current project uses SQLite for user storage.	May require a more scalable database approach as structured user data increases.	Medium
2	Workout plans are generated using Gemini.	Higher usage may require attention to API usage, response time, and service availability.	Medium
3	Application is demonstrated through a local FastAPI deployment.	Scaling FastAPI deployment would require a suitable hosting and deployment strategy.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	FastAPI application is demonstrated locally.	Deploy the application on a cloud hosting and support multiple instances for load balancing.
2	Data Storage	SQLite is used for storing user details and generated plans.	Move to a cloud database when user and data volume grows.
3	Performance	The system generates personalized 7-day workout plans.	Optimize API responses, request handling, and application logic.
4	Security	The project includes user storage and a strength assessment.	Strengthen authentication, authorization, data protection, and privacy.

Scalability & Future Plan

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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Production-ready deployment and improved database schema	Q3 2024	Supports more reliable application usage and data storage
Phase 2	Improve performance and scalability for increased users and Gemini requests	Q4 2024	Improves response handling and supports higher load
Phase 3	Enhance security, authentication, and admin access controls	Q1 2025	Provides stronger protection for user and app data
Phase 4	Extend FitBuddy with additional fitness and personalization features.	Q2 2025	Expands the project's functionality beyond the current scope

Note: The scalability limitations and roadmap items above are proposed based on the documented FitBuddy architecture and features. Actual project constraints, priorities, timelines, and implemented upgrades should be confirmed by the team.